

Experiment No 05 :- ^(A) Conductometric titration

^(B) Determination of solubility product of BaSO_4

Theory :-

What is conductometric titration.

Include,

$$G_{\text{corr}} = G_{\text{obs}} \left[\frac{V_1 + V_2}{V_1} \right]$$

Where,

G_{corr} = corrected conductance

G_{obs} = observed conductance

V_1 = initial volume of HCl

V_2 = added volume of NaOH

$$\Lambda_m = \frac{k}{C}$$

Where,

Λ_m = molar conductivity

k = conductivity

C = concentration.

$$\Lambda^\infty = (V_+ \cdot \lambda_+^\infty) + (V_- \cdot \lambda_-^\infty)$$

Where, Λ^∞ = limiting molar conductivity

V_+ = number of cations

V_- = number of anions

λ_+^∞ = limiting ionic conductivity of cation.

λ_-^∞ = limiting ionic conductivity of anion.

$$K_{sp} = [C_+] [C_-]$$

Where, C_+ = cation concentration

C_- = anion concentration.

Part A: Observations

Table No 01: Observations for strong acid - base titration

V_{NaOH} (± 0.05 mL)	G_{obs} ($\pm 0.1 \mu\text{S cm}^{-1}$)	$\frac{V_1 + V_2}{V_1}$	G_{corr} ($\pm 0.1 \mu\text{S cm}^{-1}$)
0.00			
1.00			
2.00			

Table No 02: Observations for titration of an acid mixture

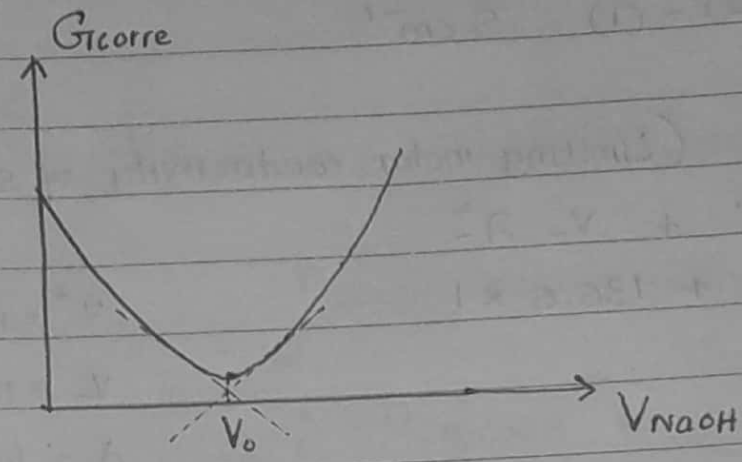
V_{NaOH} (± 0.05 mL)	G_{obs} ($\pm 0.1 \mu\text{S cm}^{-1}$)	$\frac{V_1 + V_2}{V_1}$	G_{corr} ($\pm 0.1 \mu\text{S cm}^{-1}$)
0.00			
1.00			
2.00			

Part B:-

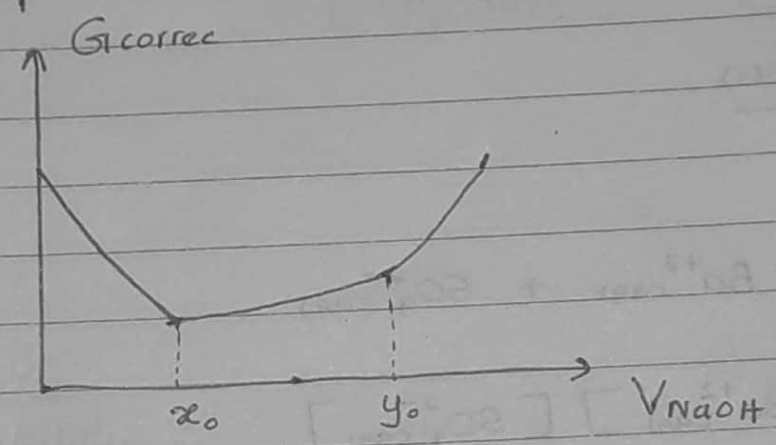
Table No 03: conductivities of water and BaSO_4

Sample	conductivity ($\pm 0.001 / \mu\text{S cm}^{-1}$)
distilled water	(a)
BaSO_4 solution.	(b)

Graph 01. : corrected conductance V_s V_{NaOH} for strong acid.



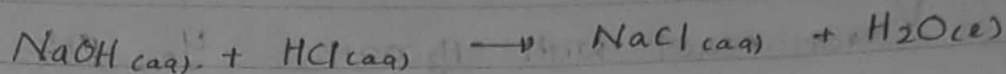
Graph 02 : corrected conductance V_s V_{NaOH} for acid mixture.



x_0 = for strong acid
 y_0 = for acid mixture

Calculations :-

Part A :-



for strong acid ;

$$V_{\text{NaOH}} \text{ react with HCl} = V_0$$

$$C_{\text{NaOH}} = 0.25 \text{ mol dm}^{-3}$$

$$n_{\text{NaOH}} = 0.25 \text{ mol dm}^{-3} \times V_0 \times 10^{-3} \text{ dm}^3$$

$$= \frac{0.25 V_0 \text{ mol}}{1000}$$

$$\therefore n_{\text{HCl}} = \frac{0.25 V_0 \text{ mol}}{1000} \quad (\because n_{\text{HCl}} = n_{\text{NaOH}})$$

$$[\text{HCl}] = \frac{0.25 V_0 \text{ mol}}{1000}$$

$$50 \times 10^{-3} \text{ dm}^3$$

$$= (x) //$$

for acid mixture ;

$$V_{\text{NaOH}} \text{ react with HCl} = x_0$$

$$[\text{NaOH}] = 0.25 \text{ mol dm}^{-3}$$

$$n_{\text{NaOH}} = 0.25 \text{ mol dm}^{-3} \times \frac{x_0}{1000} \text{ dm}^3$$

$$\therefore n_{\text{HCl}} = \frac{0.25}{1000} \times x_0 \text{ mol}$$

$$V_{\text{HCl}} = \frac{(0.25 x_0 / 1000) \text{ mol}}{(x) \text{ mol dm}^{-3}} = (V) \text{ dm}^3$$

Then $V_{\text{CH}_3\text{COOH}} = \frac{(50 - V)}{1000} \text{ dm}^3 = V'$

V_{NaOH} react with $\text{CH}_3\text{COOH} = y_0 \text{ mL}$

n_{NaOH} react with $\text{CH}_3\text{COOH} = y_0 \times 10^{-3} \times 0.25$

$\therefore n_{\text{CH}_3\text{COOH}} = \frac{0.25 y_0}{1000} \text{ mol}$

$[\text{CH}_3\text{COOH}] = \frac{0.25 y_0}{1000} \text{ mol} / V' \text{ dm}^3$

//

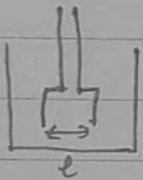
Part B

Calculations:-

Calculation for conductivity of distilled water at $T^\circ\text{C}$

conductance $\rightarrow$ conductivity

$G = \frac{kA}{l}$



$k = \frac{Gl}{A}$

$\frac{l}{A} = 1 \text{ cm}^{-1}$

$k = a \times 1 \text{ cm}^{-1} \text{ — (1)}$

Calculation for conductivity of BaSO_4 at $T^\circ\text{C}$

$G = \frac{kA}{l}$

$k = \frac{Gl}{A}$

$\frac{l}{A} = 1 \text{ cm}^{-1}$

$k = b \times 1 \text{ cm}^{-1} \text{ — (2)}$

$$K_{BaSO_4(s)} = K_{BaSO_4(aq)} - K_{H_2O}$$

$$= (2) - (1) \quad S \text{ cm}^{-1}$$

$$\Lambda_{salt} = \Lambda_{salt}^{\infty} \quad (\text{Limiting molar conductivity of salt})$$

$$\begin{aligned} \text{(molar conductivity of salt)} &= \nu^+ \lambda^+ + \nu^- \lambda^- \\ &= 1 \times 110 + 136.6 \times 1 \\ &= \end{aligned}$$

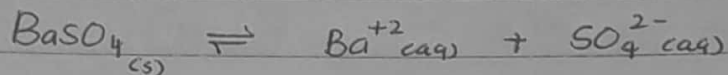
ν^+ = number of cations

ν^- = number of anions

λ = limiting ionic conductivities

$$\Lambda_{salt} = \frac{K_{BaSO_4(s)}}{C}$$

$$C = \frac{K_{BaSO_4(s)}}{\Lambda_{salt}}$$



$$\begin{aligned} \text{Solubility product } K_{sp} &= [Ba^{+2}(aq)] [SO_4^{2-}(aq)] \\ \text{constant} &= C \times C \\ &= C^2 \\ &= \end{aligned}$$

Conclusion :-

concentration of HCl

concentration of CH_3COOH

solubility product K_{sp} of $BaSO_4$

Discussion :-

- Discuss the shapes of 2 graphs
- Errors
- Advantages and disadvantage of conductometric titrations.
- Applications of conductometric titrations.